

$$\mathbb{A}_{\mu, st}(\mathbf{k}) = \langle u_s^W | i \partial_\mu | u_t^W \rangle \quad [\text{AA}], \quad (16)$$

$$\mathbb{A}_{\mu, st}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{b}_{st}(\mathbf{R})} X_{\mu, st}(\mathbf{R}), \quad [\text{ft-aa}] \quad (18)$$

$$X_{\mu, st}(\mathbf{R}) = \langle 0s | \hat{r}_\mu - \tau_{s\mu} | \mathbf{R}t \rangle, \quad [\text{r-real}] \quad \leftarrow \text{For Cartesian} \quad (11)$$

$$\mathbb{A}_{\beta, st}(\underline{k}) = \sum_{\underline{R}} e^{i\underline{k} \cdot \underline{b}_{st}(\underline{R})} X_{\beta, st}(\underline{R}) \quad ?? \quad \text{Berry connection in } \beta$$

Find expression for $X_{\beta, st}(\underline{R})$?

$$|u_s^W\rangle = \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R} - \tau_s)} |\mathbf{R}s\rangle \quad [\text{wft}]$$

$\uparrow$
 τ_s can act here
 $\leftarrow$ where

$$\mathbb{A}_{\beta, st}(\underline{k}) = \frac{1}{N} \sum_{\mathbf{R}, \mathbf{R}'} e^{i\underline{k} \cdot (\underline{r} - \underline{R}' - \tau_s)} e^{-i\underline{k} \cdot (\underline{r} - \underline{R} - \tau_t)} \left[\langle \mathbf{R}'s | i \partial_\beta | \mathbf{R}t \rangle - \delta_{\mathbf{R}\mathbf{R}'} \delta_{st} \underline{k} \cdot \partial_\beta \tau_t \right]$$

$\downarrow \underline{R} = \underline{R}' - \underline{R}'$

$$\mathbb{A}_{\beta, st}(\underline{k}) = \sum_{\underline{R}} e^{i\underline{k} \cdot \underline{b}_{st}(\underline{R})} \left[\underbrace{\langle 0s | i \partial_\beta | \underline{R}t \rangle}_{\leftarrow \text{Define as } X_{\beta, st}(\underline{R})} - \delta_{0\underline{R}} \delta_{st} \underline{k} \cdot \partial_\beta \tau_t \right]$$

$\leftarrow k\text{-dep, cannot be absorbed into } X$

$$\mathbb{A}_{\beta, st}(\underline{k}) = \sum_{\underline{R}} e^{i\underline{k} \cdot \underline{b}_{st}(\underline{R})} X_{\beta, st}(\underline{R}) - \delta_{st} \underline{k} \cdot (\partial_\beta \tau_t)$$

$\uparrow$
 $\partial_\mu \mathbb{A}_{\beta, st}$ acts in 2 places $\uparrow$

also need $\partial_\beta \mathbb{A}_{\mu, st}$

This is only for purely external part

Evaluate $X_{\beta, st}(\underline{R}) = \langle 0s^0 | i \partial_\beta | \underline{R}t^0 \rangle \quad \leftarrow \beta \text{ arg put up here}$

$$= 0, \text{ if } 0s = \underline{R}t, \text{ I think}$$

$$= i \frac{\langle 0s^0 | \underline{R}t^0 \rangle}{\underline{k} \cdot \underline{b}} \quad \text{, if } 0s \neq \underline{R}t \quad (\text{finite diff})$$

Has to be calculated in 1st - princ calc.

$\uparrow$
 2nd term can be eliminated if you insist on using undisplaced atomic coords for $\underline{I}$.
 What does Pyth TB use, when importing from VBO?